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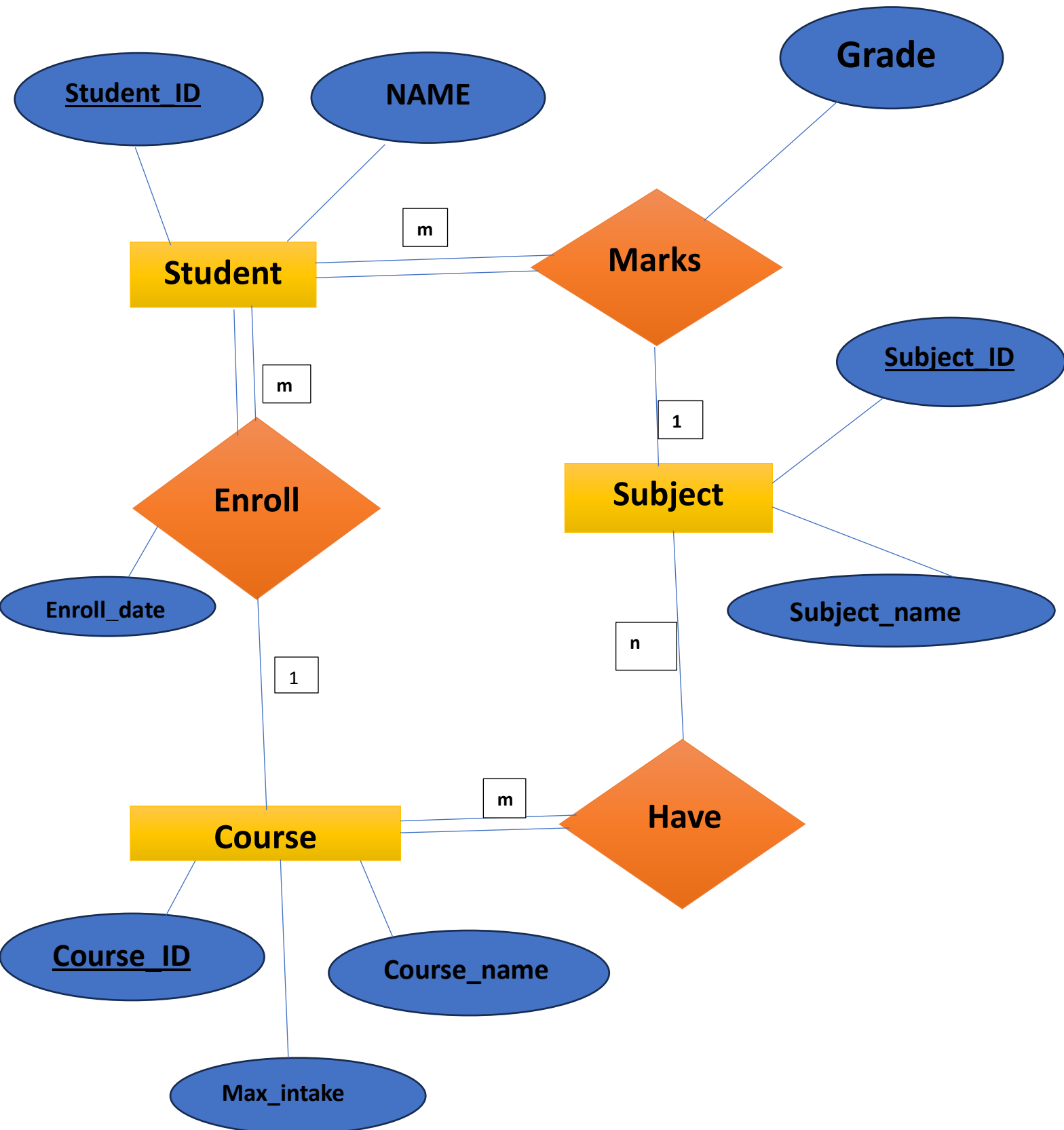
YEAR :2ND

ASSIGNMENT NO.-3

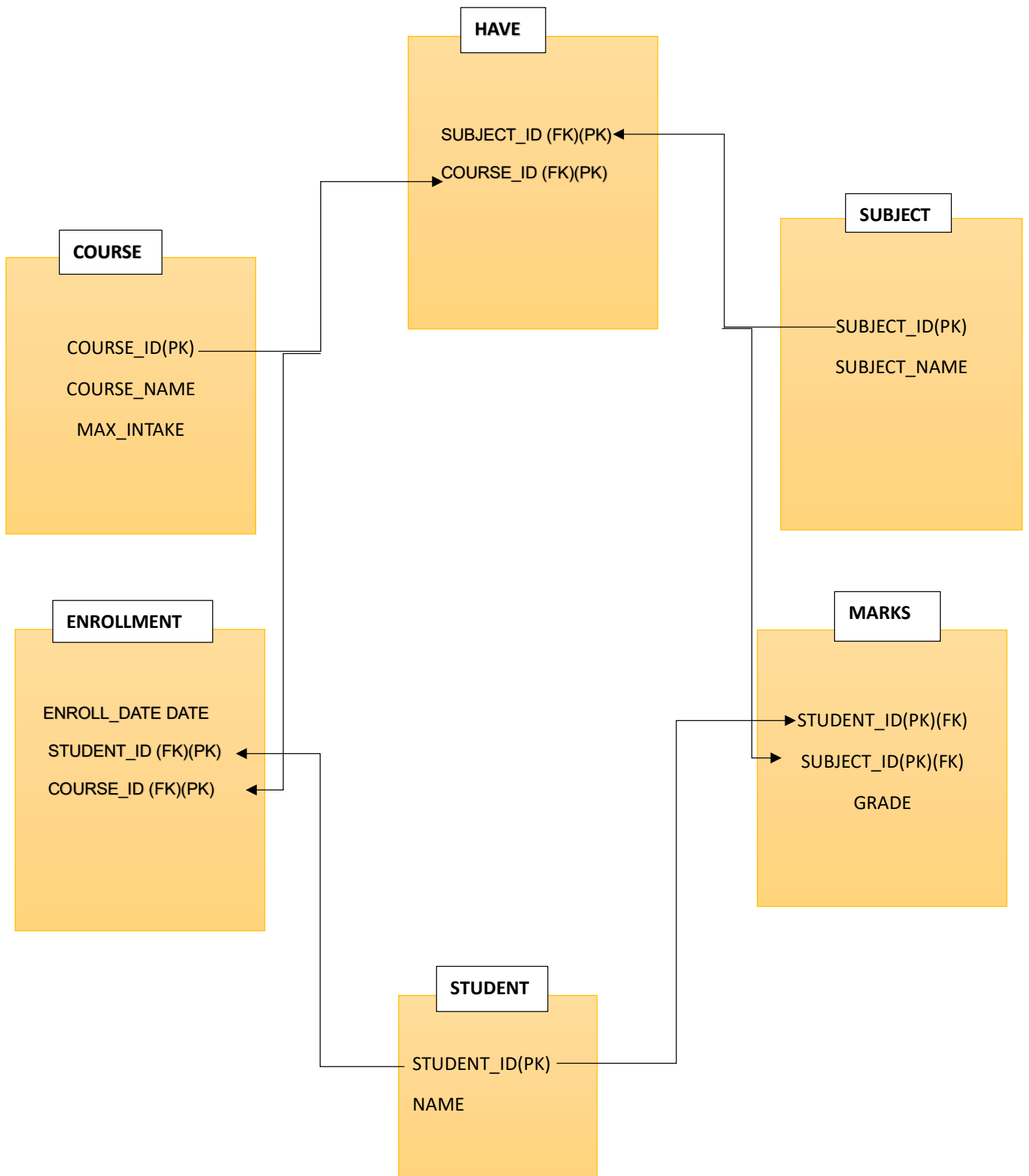
SUBJECT-DBMS

ASSIGNMENT-3

E-R DIAGRAM :



Database Schema:



TABLES:

Student Table

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(100) );
```

Course Table

```
CREATE TABLE Course (  
    Course_ID INT PRIMARY KEY,  
    Course_Name VARCHAR(100),  
    Max_Intake INT CHECK (Max_Intake > 0) );
```

Subject Table

```
CREATE TABLE Subject (  
    Subject_ID INT PRIMARY KEY,  
    Subject_Name VARCHAR(100) );
```

Enrollment Table

```
CREATE TABLE Enrollment (  
    Enroll_Date DATE,  
    Student_ID INT,  
    Course_ID INT,  
    PRIMARY KEY (Student_ID, Course_ID), -- Composite Primary Key  
    FOREIGN KEY (Student_ID) REFERENCES Student(Student_ID),  
    FOREIGN KEY (Course_ID) REFERENCES Course(Course_ID));
```

HAVE Table (to relate Courses and Subjects)

```
CREATE TABLE HAVE (  
    Course_ID INT,  
    Subject_ID INT,  
    PRIMARY KEY (Course_ID, Subject_ID), -- Composite Primary Key  
    FOREIGN KEY (Course_ID) REFERENCES Course(Course_ID),  
    FOREIGN KEY (Subject_ID) REFERENCES Subject(Subject_ID));
```

Marks Table (relating only Student and Subject)

```
CREATE TABLE Marks (  
    Student_ID INT,  
    Subject_ID INT,  
    Grade CHAR(1) CHECK (Grade IN ('A', 'B', 'C', 'D', 'F')),  
    PRIMARY KEY (Student_ID, Subject_ID),  
    FOREIGN KEY (Student_ID) REFERENCES Student(Student_ID),  
    FOREIGN KEY (Subject_ID) REFERENCES Subject(Subject_ID) );
```

Q1. Insert at least five tuples in each table.

Query:

Insert into Student Table:

```
INSERT INTO Student (Student_ID, Name) VALUES (18065, 'Alice Johnson');  
INSERT INTO Student (Student_ID, Name) VALUES (18066, 'Bob Smith');  
INSERT INTO Student (Student_ID, Name) VALUES (18067, 'Charlie Davis');  
INSERT INTO Student (Student_ID, Name) VALUES (18068, 'Diana Brown');  
INSERT INTO Student (Student_ID, Name) VALUES (18069, 'Evan Clark');  
INSERT INTO Student (Student_ID, Name) VALUES (18070, 'Fiona White');  
INSERT INTO Student (Student_ID, Name) VALUES (18071, 'George Lewis');  
INSERT INTO Student (Student_ID, Name) VALUES (18072, 'Hannah Scott');
```

Insert into Subject Table:

```
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (1, 'AI');  
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (2, 'DBMS');  
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (3, 'DSA');  
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (4, 'ML');  
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (5, 'Computer Networks');  
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (6, 'Operating Systems');  
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (7, 'Software Engineering');
```

```
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (8, 'Cybersecurity');
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (9, 'Cloud Computing');
INSERT INTO Subject (Subject_ID, Subject_Name) VALUES (10, 'Web Development');
```

Insert into Course Table:

```
INSERT INTO Course (Course_ID, Course_Name, Max_Intake) VALUES (10, 'Computer
Science and Engineering', 60);
INSERT INTO Course (Course_ID, Course_Name, Max_Intake) VALUES (20, 'Information
Technology', 50);
INSERT INTO Course (Course_ID, Course_Name, Max_Intake) VALUES (30, 'Electrical
Engineering', 40);
INSERT INTO Course (Course_ID, Course_Name, Max_Intake) VALUES (40,
'Instrumentation Engineering', 55);
INSERT INTO Course (Course_ID, Course_Name, Max_Intake) VALUES (50, 'Mechanical
Engineering', 45);
INSERT INTO Course (Course_ID, Course_Name, Max_Intake) VALUES (60, 'Civil
Engineering', 30);
INSERT INTO Course (Course_ID, Course_Name, Max_Intake) VALUES (70, 'Chemical
Engineering', 40);
INSERT INTO Course (Course_ID, Course_Name, Max_Intake) VALUES (80, 'Aerospace
Engineering', 35);
INSERT INTO Course (Course_ID, Course_Name, Max_Intake) VALUES (90, 'Philosophy',
40);
```

Insert into Enrollment Table

-- Student 18065

```
INSERT INTO Enrollment (Enroll_Date, Student_ID, Course_ID) VALUES
('2024-09-01', 18065, 10),
('2024-09-01', 18065, 20),
('2024-09-01', 18065, 30),
('2024-09-01', 18065, 40),
('2024-09-01', 18065, 50),
```

-- Student 18066

('2024-09-01', 18066, 10),
('2024-09-01', 18066, 20),
('2024-09-01', 18066, 30),
('2024-09-01', 18066, 40),
('2024-09-01', 18066, 50),
('2024-09-01', 18066, 60),

-- Student 18067

('2024-09-01', 18067, 20),
('2024-09-01', 18067, 30),
('2024-09-01', 18067, 40),
('2024-09-01', 18067, 50),
('2024-09-01', 18067, 60),

-- Student 18068

('2024-09-01', 18068, 30),
('2024-09-01', 18068, 40),
('2024-09-01', 18068, 50),
('2024-09-01', 18068, 60),
('2024-09-01', 18068, 70),
('2024-09-01', 18068, 80),

-- Student 18069

('2024-09-01', 18069, 40),
('2024-09-01', 18069, 50),
('2024-09-01', 18069, 60),
('2024-09-01', 18069, 70),
('2024-09-01', 18069, 80),

-- Student 18070

('2024-09-01', 18070, 50),

('2024-09-01', 18070, 60),

('2024-09-01', 18070, 70),

('2024-09-01', 18070, 80),

('2024-09-01', 18070, 10),

-- Student 18071

('2024-09-01', 18071, 60),

('2024-09-01', 18071, 70),

('2024-09-01', 18071, 80),

('2024-09-01', 18071, 50),

('2024-09-01', 18071, 30),

-- Student 18072

('2024-09-01', 18072, 70),

('2024-09-01', 18072, 80),

('2024-09-01', 18072, 50),

('2024-09-01', 18072, 30),

('2024-09-01', 18072, 10);

INSERT INTO HAVE

INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (10, 1);

INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (10, 2);

INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (10, 3);

INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (10, 4);

INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (10, 5);

INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (10, 6);

INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (10, 7);

INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (20, 8);

```
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (20, 9);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (20, 10);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (20, 1);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (20, 2);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (20, 3);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (20, 4);
```

```
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (30, 5);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (30, 6);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (30, 7);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (30, 8);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (30, 9);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (30, 10);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (30, 1);
```

```
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (40, 2);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (40, 3);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (40, 4);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (40, 5);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (40, 6);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (40, 7);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (40, 8);
```

```
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (50, 9);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (50, 10);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (50, 1);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (50, 2);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (50, 3);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (50, 4);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (50, 5);
```

```
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (60, 6);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (60, 7);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (60, 8);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (60, 9);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (60, 10);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (60, 1);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (60, 2);
```

```
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (70, 3);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (70, 4);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (70, 5);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (70, 6);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (70, 7);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (70, 8);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (70, 9);
```

```
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (80, 10);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (80, 1);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (80, 2);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (80, 3);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (80, 4);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (80, 5);
INSERT INTO HAVE (Course_ID, Subject_ID) VALUES (80, 6);
```

Insert into Marks Table

-- Student 18065

```
INSERT INTO Marks (Student_ID, Subject_ID, Grade) VALUES
(18065, 1, 'A'),
```

(18065, 2, 'B'),

(18065, 3, 'A'),

(18065, 4, 'C'),

(18065, 5, 'B'),

-- Student 18066

(18066, 1, 'B'),

(18066, 2, 'A'),

(18066, 3, 'B'),

(18066, 4, 'A'),

(18066, 5, 'C'),

(18066, 6, 'B'),

-- Student 18067

(18067, 2, 'A'),

(18067, 3, 'B'),

(18067, 4, 'A'),

(18067, 5, 'B'),

(18067, 6, 'C'),

-- Student 18068

(18068, 3, 'B'),

(18068, 4, 'A'),

(18068, 5, 'C'),

(18068, 6, 'B'),

(18068, 7, 'A'),

(18068, 1, 'B'),

-- Student 18069

(18069, 4, 'C'),

(18069, 5, 'B'),
(18069, 6, 'A'),
(18069, 7, 'B'),
(18069, 1, 'A'),

-- Student 18070

(18070, 5, 'A'),
(18070, 6, 'B'),
(18070, 7, 'A'),
(18070, 1, 'C'),
(18070, 2, 'B'),

-- Student 18071

(18071, 6, 'B'),
(18071, 7, 'A'),
(18071, 1, 'A'),
(18071, 2, 'C'),
(18071, 3, 'B'),

-- Student 18072

(18072, 7, 'B'),
(18072, 1, 'A'),
(18072, 2, 'B'),
(18072, 3, 'C'),
(18072, 4, 'A');

2. At the time of creation if we forget to create a field enrollment date (ENROLL_DATE)
in ENROLL table so add the field.

Query: ALTER TABLE Enrollment ADD Enroll_Date DATE;

Output:

	Field	Type	Null	Key
▶	Enroll_Date	date	YES	
	Student_ID	int	NO	PRI
	Course_ID	int	NO	PRI

	Enroll_Date	Student_ID	Course_ID
▶	2024-09-01	18065	10
	2024-09-01	18065	20
	2024-09-01	18065	30
	2024-09-01	18065	40
	2024-09-01	18065	50
	2024-09-01	18066	10
	2024-09-01	18066	20
	2024-09-01	18066	30
	2024-09-01	18066	40
	2024-09-01	18066	50
	2024-09-01	18066	60
	2024-09-01	18067	20
	2024-09-01	18067	30
	2024-09-01	18067	40
	2024-09-01	18067	50
	2024-09-01	18067	60
	2024-09-01	18068	30
	2024-09-01	18068	40
	2024-09-01	18068	50
	2024-09-01	18068	60
	2024-09-01	18068	70
	2024-09-01	18068	80
	2024-09-01	18069	40
	2024-09-01	18069	50
	2024-09-01	18069	60

3. Course name cannot be blank, therefore add the criteria in the specific table.

Query:

ALTER TABLE Course

MODIFY Course_Name VARCHAR(100) NOT NULL;

Output:

	Field	Type	Null	Key
▶	Course_ID	int	NO	PRI
	Course_Name	varchar(100)	NO	
	Max_Intake	int	YES	

4. Find the Course which has more than 3 students.

Query:

```

SELECT c.Course_Name, COUNT(e.Student_ID) AS Number_of_Students
FROM Enrollment e
JOIN Course c ON e.Course_ID = c.Course_ID
GROUP BY c.Course_ID, c.Course_Name
HAVING COUNT(e.Student_ID) > 3;

```

Output:

	Course_Name	Number_of_Students
►	Computer Science and Engineering	4
	Electrical Engineering	6
	Instrumentation Engineering	5
	Mechanical Engineering	8
	Civil Engineering	6
	Chemical Engineering	5
	Aerospace Engineering	5

5. Give the details of a STUDENT with all Subjects and Grade where he/she enroll (Enter the sid value as input).

Query:

```

SELECT
    s.Student_ID,
    s.Name AS Student_Name,
    sub.Subject_ID,
    sub.Subject_Name,
    m.Grade
FROM
    Student s
JOIN
    Marks m ON s.Student_ID = m.Student_ID
JOIN
    Subject sub ON m.Subject_ID = sub.Subject_ID
WHERE

```

s.Student_ID = 18070;

Output:

Result Grid					
		Filter Rows:		Export:	Wrap Cell Co
	Student_ID	Student_Name	Subject_ID	Subject_Name	Grade
▶	18070	Fiona White	1	AI	C
	18070	Fiona White	2	DBMS	B
	18070	Fiona White	5	Computer Networks	A
	18070	Fiona White	6	Operating Systems	B
	18070	Fiona White	7	Software Engineering	A

6. Display the course where the maximum number of students enrolls.

Query:

```
SELECT c.Course_Name, COUNT(e.Student_ID) AS Number_of_Students
FROM Enrollment e
JOIN Course c ON e.Course_ID = c.Course_ID
GROUP BY c.Course_ID, c.Course_Name
HAVING COUNT(e.Student_ID) = (
    SELECT MAX(Student_Count)
    FROM (
        SELECT COUNT(e.Student_ID) AS Student_Count
        FROM Enrollment e
        GROUP BY e.Course_ID
    ) AS Subquery
);
```

Output:

	Course_Name	Number_of_Students
▶	Mechanical Engineering	8

7. Find out the course where no student is enrolled.

Query:

```
SELECT c.Course_Name
FROM Course c
```

LEFT JOIN Enrollment e ON c.Course_ID = e.Course_ID

WHERE e.Student_ID IS NULL;

Output:

	Course_Name
►	Philosophy

8. Delete Course no 30 from COURSE table.

Query:

DELETE FROM Course

WHERE Course_ID = 30;

9. Rename the COURSE table as DEPARTMENT.

Query:

RENAME TABLE Course TO Department;

Output:

After ->SELECT* FROM DEPARTMENT ;

	Course_ID	Course_Name	Max_Intake
►	10	Computer Science and Engineering	60
	20	Information Engineering and Electronics	50
	30	Electrical Engineering	40
	40	Instrumentation Engineering	55
	50	Mechanical Engineering	45
	60	Civil Engineering	30
	70	Chemical Engineering	40
	80	Aerospace Engineering	35
	90	Philosophy	40

10. Change the Marks Grade of Student “A” to “B” who is Enroll in the subject DBMS.

Query :

UPDATE Marks

SET Grade = 'B'

WHERE Grade = 'A'

```
AND Subject_ID = (  
    SELECT Subject_ID  
    FROM Subject  
    WHERE Subject_Name = 'DBMS');
```

11.Delete the record of the student who is enrolled in the course 'IT'.

Query:

```
DELETE FROM Student  
WHERE Student_ID IN (  
    SELECT DISTINCT e.Student_ID  
    FROM Enrollment e  
    JOIN Department d ON e.Course_ID = d.Course_ID  
    WHERE d.Course_Name = 'Information Technology'  
);
```

12. Change the enroll date to '16-08-2018' whose student id is 18069 (first convert the date into the default format).

Query:

```
UPDATE Enrollment  
SET Enroll_Date = '2018-08-16'  
WHERE Student_ID = 18069;
```

Output:

2018-08-16	18069	40
2018-08-16	18069	50
2018-08-16	18069	60
2018-08-16	18069	70
2018-08-16	18069	80
2024-09-01	18070	10
2024-09-01	18070	50